CHÂTELET'S THEOREM IN SYNTHETIC ALGEBRAIC GEOMETRY: A CHECKED PROOF

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Abstract. We present the proof of Châtelet's theorem in synthetic algebraic geometry that has been checked in the proof assistant Unit: a Severi–Brauer variety with a point is projective space. The proof follows the note *Châtelet's theorem in synthetic algebraic geometry* (sevb), with three changes. For the étale topology it is checked from the axioms of synthetic algebraic geometry and a left exact sheafification; the descent of finite freeness that it needs is proved in a companion note. Right ideals are required to be free *direct summands*. All "same dimension, hence equal" steps are replaced by one fact on square matrices over a local ring, proved without determinants. Compactness of the spectrum of $R[X]/(f)$, for f monic, is proved algebraically. We also present the part of sevb's Section 1 that has been checked: for the topology generated by roots of monic polynomials, R is a sheaf and so is every scheme. The last section lists, for each statement, the corresponding checked definition.

1. Setting and statement

R is the ring of synthetic algebraic geometry: it is local (that is, $1 \neq 0$, and if $a + b$ is invertible then a or b is invertible), and it satisfies synthetic quasi-coherence and Zariski-local choice [1]. We work in univalent type theory; "merely" means inside the propositional truncation $\| - \|$. Fix $n \geq 1$ and put $N = n + 1$.

Projective space $\mathbb{P}^n$ is the type of vectors $x \in R^N$ with some coordinate invertible, modulo $x \sim ux$ for u a unit. $M_N(R)$ is the algebra of $N \times N$ matrices, E_{ij} the elementary matrices, $\mathrm{PGL}_N(R)$ the invertible linear maps of R^N modulo unit scalars.

The topology. A topology T is given by a family of propositions G_i $(i \in I)$. A type X is a *sheaf* if for every i the constant map $X \to X^{G_i}$, $x \mapsto (_ \mapsto x)$, is an equivalence. We assume:

(H1) R is a sheaf.
(H2) For every type A there is a proposition $\|A\|_T$ that is a sheaf, with a map $A \to \|A\|_T$, such that every map from A to a proposition that is a sheaf factors through $\|A\|_T$.
(H3) (Lexness) The type of sheaves Σ_X (X is a sheaf) is a sheaf.
(H4) (Descent) For an algebra A whose carrier is a sheaf and a right ideal I of A (Definition 5.1), if I has a basis T-locally then it merely has a basis.

For the étale topology of Section 8, (H1) is proved there, and (H4) is proved in [3] for the algebras that are used: those with $\|A = M_N(R)\|_T$, including $M_N(R)$ itself.

Definition 1.1. AZ_n is the type of R-algebras A whose carrier is a sheaf and such that $\|A = M_N(R)\|_T$. SB_n is the type of sheaves X such that $\|X = \mathbb{P}^n\|_T$.

Theorem 1.2 (Châtelet)**.** *For $X : \mathrm{SB}_n$, we have $\|X\| \to \|X = \mathbb{P}^n\|$.*

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The proof has three parts. The map $A \mapsto \mathrm{RI}(A)$, right ideals of rank N, is an equivalence $\mathrm{AZ}_n \simeq \mathrm{SB}_n$ (Proposition 6.3); both types are deloopings of $\mathrm{PGL}_N(R)$, since $\mathrm{Aut}(M_N(R)) = \mathrm{PGL}_N(R) = \mathrm{Aut}(\mathbb{P}^n)$. A point of $\mathrm{RI}(A)$ is a right ideal I, and A acts on I by an isomorphism $A \simeq M_N(R)$ (Lemma 7.1). Hence $X = \mathrm{RI}(A) = \mathrm{RI}(M_N(R)) = \mathbb{P}^n$ (Lemma 5.3).

2. Sheaves

Lemma 2.1. *Sheaves are closed under* Π *(over any type),* Σ*, identity types and retracts. A proposition* P *is a sheaf if and only if every map* $G_i \to P$ *has a value, i.e.* $(G_i \to P) \to P$ *for all* i*.*

Proof. For Π the inverse is pointwise. For $\Sigma_{a:A}B(a)$, the constant map is the composite

$$\Sigma_a B(a) \to \Sigma_a B(a)^G \to \Sigma_{\alpha:G\to A}\Pi_g B(\alpha g) \cong (G \to \Sigma_a B(a)),$$

of a fibrewise equivalence, the base change of a family along the equivalence $A \to A^G$ (an equivalence by induction on equivalences), and a canonical isomorphism. For $x, y : X$ the constant map $(x = y) \to (G \to x = y)$ is ap of the equivalence $X \to X^G$ followed by function extensionality. For a retract $s : B \to A$, $r : A \to B$, $rs = 1$, with A a sheaf, the inverse of the constant map of B sends $h : G \to B$ to $r(a)$, where a is the value of $s \circ h$. For a proposition both composites of the two maps are equalities between elements of propositions. □

Lemma 2.2. *The type* Prop_T *of propositions* P *with sections* $(G_i \to P) \to P$ *is a sheaf, and two elements of it with the same inhabitants are equal.*

Proof. By Lemma 2.1 it is a retract of $\Sigma_{X\ \mathrm{sheaf}}(X \text{ is a proposition})$. This is a Σ over the type of sheaves, a sheaf by (H3), of a family of sheaves, since "X is a proposition" is a Π of identity types. The second claim is univalence; the sections form a proposition. □

Lemma 2.3 (Delooping, sevb 3.1.2)**.** *Let* A, B *be sheaves with base points* a_0, b_0*, each* connected *(*$\|x = y\|_T$ *for all* x, y*), and* $f : A \to B$ *with* $p : f(a_0) = b_0$*. If* $\Omega f : (a_0 = a_0) \to (b_0 = b_0)$*,* $q \mapsto p^{-1} \cdot f(q) \cdot p$*, is an equivalence, then* f *is an equivalence.*

Proof. "$\mathrm{ap}_f : (x = y) \to (fx = fy)$ has contractible fibres" is a proposition and a sheaf (Lemma 2.1). By connectedness we may take $x = y = a_0$, where ap_f is Ωf conjugated by p. So the fibres of f are propositions; they are sheaves, so by connectedness of B each is inhabited, as the fibre over b_0 is. □

3. Square matrices over a local ring

Lemma 3.1 (K1)**.** *If* $\sum_i \lambda_i v_i = 1$ *in* R*, then merely some* v_i *is invertible.*

Proof. By locality some $\lambda_i v_i$ is invertible, hence v_i is. □

Lemma 3.2 (K2)**.** *If* $L, M \in M_m(R)$ *and* $LM = 1$ *then* $ML = 1$*.*

Proof. Induction on m; the case $m = 0$ is trivial. Let t be the last index and c the last column of M. Since $\sum_k L_{tk}c_k = 1$, by locality c_t is invertible or merely some c_k, $k \neq t$, is. In the second case $E = 1 + aE_{tk}$, with $a = (1 - c_t)c_k^{-1}$, satisfies $(1 - aE_{tk})E = 1$ and $(EM)_{tt} = 1$. So merely there is P with a left inverse and $(PM)_{tt}$ invertible. Scaling row t by $(PM)_{tt}^{-1}$ and clearing the rest of the last column is a matrix F with an explicit left inverse. So $Q = FP$ has a left inverse Q' and the last column of QM is e_t. The claim is a proposition, and it is invariant under $(L, M) \mapsto (LQ', QM)$: if $(QM)(LQ') = 1$ then $ML = Q'Q\,ML\,Q'Q = Q'(QM\,LQ')Q = 1$. So we may assume $M = \begin{pmatrix} A & 0 \\ r & 1 \end{pmatrix}$, and write

$L = \begin{pmatrix} B & s \\ u & d \end{pmatrix}$. Then $LM = 1$ says $s = 0$, $d = 1$, $BA = 1$ and $uA = -r$. By induction $AB = 1$, so $rB = -uAB = -u$, and $ML = \begin{pmatrix} AB & As \\ rB + u & rs + d \end{pmatrix} = 1$. □

4. Skolem–Noether

Proposition 4.1 (Skolem–Noether)**.** *Every unital algebra endomorphism φ of $M_N(R)$ is merely of the form $X \mapsto WXW^{-1}$.*

Proof. Put $e_{ab} = \varphi(E_{ab})$, so $e_{ab}e_{cd} = \delta_{bc}e_{ad}$ and $\sum_j e_{jj} = 1$. Fix an index t. Since $\sum_j (e_{jj})_{tt} = 1$, by Lemma 3.1 merely $c = (e_{jj})_{tt}$ is invertible for some j. Put $W_{xy} = (e_{yj})_{xt}$ and $L_{xy} = c^{-1}(e_{jx})_{ty}$. Then $(LW)_{xy} = c^{-1}(e_{jx}e_{yj})_{tt} = \delta_{xy}$, so $WL = 1$ by Lemma 3.2. Both $\varphi(X)W$ and WX have entries $\varphi(XE_{yj})_{xt}$, since $XE_{yj} = \sum_z X_{zy}E_{zj}$. Hence $\varphi(X) = WXL$. □

Corollary 4.2. *The map $\alpha : \mathrm{PGL}_N(R) \to \mathrm{Aut}(M_N(R))$, $[G] \mapsto (X \mapsto GXG^{-1})$, is an equivalence.*

Proof. It is well defined since scalars cancel. It is surjective by Proposition 4.1. If $GXG^{-1} = G'XG'^{-1}$ for all X, then $P = G'^{-1}G$ commutes with every E_{ij}, so $P = \mu \cdot 1$ with μ a unit, and $[G] = [G']$. □

We also use the equivalence $\beta : \mathrm{PGL}_N(R) \to \mathrm{Aut}(\mathbb{P}^n)$, the action of $\mathrm{PGL}_N(R)$ on $\mathbb{P}^n$: this is the description of the automorphisms of projective space of [2] (Corollary 5.6 there), checked earlier for every $n \geq 1$.

5. Right ideals and the map δ

Definition 5.1. Let A be an R-algebra. A *right ideal* of A is a map $I : A \to \mathrm{Prop}_T$ closed under 0, +, scalars and right multiplication by A. A *basis* of I is a linear bijection $f : R^N \to I$ together with a linear map $g : A \to R^N$, defined on all of A, with $gf = 1$. $\mathrm{RI}(A)$ is the type of right ideals I with $\|I \text{ has a basis}\|_T$.

Remark 5.2. The left inverse g makes I a free direct summand of A, as in sevb's Grassmannian. It is needed: for t a non-invertible non-zero-divisor, $(t) \subseteq R$ is a right ideal of $M_1(R) = R$ that is free of rank one, but it is not of the form $\delta(q)$ below.

$\mathrm{RI}(A)$ is a sheaf by Lemmas 2.1 and 2.2, for any A.

For $x \in R^N$ and $X \in M_N(R)$ write $\mathrm{Col}_x(X)$ for: $X_{ij}x_k = X_{kj}x_i$ for all i, j, k, i.e. every column of X is proportional to x. If x_k is invertible, $\mathrm{Col}_x(X)$ holds if and only if $X = xy^T$ for some y, and then $y = x_k^{-1}X_{k-}$.

Lemma 5.3 (sevb 2.4.2)**.** *The map $\delta : \mathbb{P}^n \to \mathrm{RI}(M_N(R))$, given by*

$$\delta(q) = \{X \mid \mathrm{Col}_x(X) \text{ for every representative } x \text{ of } q\},$$

is an equivalence. On a representative, $\delta[x] = \{xy^T \mid y \in R^N\}$.

Proof. $\mathrm{Col}_x(X)$ is a sheaf proposition by (H1), and it is invariant under $x \mapsto ux$; so membership in $\delta(q)$ needs no choice of representative. Col_x is closed under the ideal operations, e.g. $(XY)_{ij}x_k = \sum_m X_{im}x_kY_{mj} = \sum_m X_{km}x_iY_{mj} = (XY)_{kj}x_i$. For x_k invertible, $f(y) = xy^T$ and $g(X) = x_k^{-1}X_{k-}$ form a basis of $\delta[x]$.

Injective. If $\delta(q) \subseteq \delta(q')$, with $q = [x]$, $q' = [x']$, then $x\mathbf{1}^T \in \delta[x']$ gives $x_ix'_k = x_kx'_i$, so x, x' are proportional and $q = q'$.

Surjective. Let $I : \mathrm{RI}(M_N(R))$; the fibre of δ over I is a proposition, so by (H4) we may take a basis (f, g) of I. Let t be an index and $M = f(e_t)$. Since g_t is linear, $1 = g_t(M) = \sum_{a,b} M_{ab}\, g_t(E_{ab})$, so by Lemma 3.1 merely some M_{ab} is invertible. Let c be the column b of M. Then $cy^T = M(e_b y^T) \in I$ for all y. The matrices $M'_{ij} = g(ce_j^T)_i$ and $L_{ji} = c_a^{-1} f(e_i)_{aj}$ satisfy $LM' = 1$, since $\sum_i f(e_i)\, g(ce_{j'}^T)_i = f(g(ce_{j'}^T)) = ce_{j'}^T$. By Lemma 3.2, $M'L = 1$, which says $g(cr_i^T) = e_i$ with $(r_i)_j = L_{ji}$. Since $cr_i^T \in I$ and fg is the identity on I, $f(e_i) = cr_i^T$. So every element of I is of the form cy^T, and $I = \delta[c]$. □

In particular $\mathbb{P}^n$ is a sheaf. If $A : \mathrm{AZ}_n$ then $\mathrm{RI}(A) : \mathrm{SB}_n$ (sevb 2.4.3): "$\mathrm{RI}(A) = \mathbb{P}^n$" follows from $A = M_N(R)$ and the lemma, so $\|A = M_N(R)\|_T$ gives $\|\mathrm{RI}(A) = \mathbb{P}^n\|_T$.

6. The equivalence $\mathrm{AZ}_n \simeq \mathrm{SB}_n$

An algebra isomorphism $\varphi : A \simeq B$ acts on right ideals by $\varphi[I] = \{y \mid \varphi^{-1}(y) \in I\}$, with basis $(\varphi f, g\varphi^{-1})$. By the structure identity principle, φ gives a path $\mathrm{sip}(\varphi) : A = B$.

Lemma 6.1 (transport of structure)**.** *Transport along* $\mathrm{ap}_{\mathrm{RI}}(\mathrm{sip}(\varphi))$ *is* $I \mapsto \varphi[I]$.

Proof. By induction on isomorphisms it is enough to take $\varphi = 1$, where both sides are I: $\mathrm{sip}(1)$ is reflexivity, and $1[I]$ differs from I only in proofs of propositions. □

Lemma 6.2 (the triangle)**.** *For* $p \in \mathrm{PGL}_N(R)$ *and* $q \in \mathbb{P}^n$, $\alpha(p)[\delta(q)] = \delta(\beta(p)q)$.

Proof. Both sides are right ideals, so it is enough to compare members, and we may take $p = [G]$, $q = [x]$. With $H = G^{-1}$, $Y \in \alpha(p)[\delta[x]]$ means $\mathrm{Col}_x(HYG)$. The identity $P(xr^T)Q = (Px)(Q^T r)^T$ shows that $\mathrm{Col}_x(Z)$ implies $\mathrm{Col}_{Px}(PZQ)$, when x has an invertible coordinate. Applied with (G, H) and with (H, G), this gives $\mathrm{Col}_x(HYG) \iff \mathrm{Col}_{Gx}(Y)$. □

Proposition 6.3 (sevb 3.3.1)**.** $\mathrm{RI} : \mathrm{AZ}_n \to \mathrm{SB}_n$ *is an equivalence.*

Proof. Sheaves. SB_n is a Σ over the type of sheaves (H3) of sheaves. AZ_n is a retract of a Σ over the type of sheaves X of the algebra structures on X and of $\|- = M_N(R)\|_T$. Algebra structures on a sheaf form a sheaf: the operations are maps into X, and the laws, as well as "X is a set", are built from identity types of X.

Connected and pointed. $\|A = M_N(R)\|_T$ and $\|A' = M_N(R)\|_T$ give $\|A = A'\|_T$, the other components being propositions; likewise for SB_n. The base points are $M_N(R)$ and $\mathbb{P}^n$, and $\mathrm{RI}(M_N(R)) = \mathbb{P}^n$ by Lemma 5.3.

Loops. Since the other components are propositions, loops at $M_N(R)$ in AZ_n are loops in algebras, hence automorphisms of $M_N(R)$ by the structure identity principle, hence $\mathrm{PGL}_N(R)$ by α. This gives an equivalence $h : \mathrm{PGL}_N(R) \simeq \Omega(\mathrm{AZ}_n, M_N(R))$, $p \mapsto \mathrm{sip}(\alpha p)$. Loops at $\mathbb{P}^n$ in SB_n are by univalence the automorphisms of $\mathbb{P}^n$, via $k(\ell) =$ transport along ℓ. For $p \in \mathrm{PGL}_N(R)$, $\Omega\mathrm{RI}(h\, p)$ is the loop $\mathrm{ap}_{\mathrm{RI}}(\mathrm{sip}(\alpha p))$ conjugated by the path $\mathrm{RI}(M_N(R)) = \mathbb{P}^n$ of Lemma 5.3, whose transport is δ^{-1}. By Lemmas 6.1 and 6.2, transport along it is $q \mapsto \delta^{-1}(\alpha(p)[\delta(q)]) = \beta(p)q$. So $k \circ \Omega\mathrm{RI} \circ h = \beta$, and $\Omega\mathrm{RI}$ is an equivalence. Conclude by Lemma 2.3. □

7. Châtelet's theorem

Lemma 7.1 (sevb 3.4.1)**.** *Let* $A : \mathrm{AZ}_n$, $I : \mathrm{RI}(A)$ *and* (f, g) *a basis of* I. *Then* $\sigma(a)_{ij} = g(f(e_i)\, a)_j$ *is an algebra isomorphism* $A \simeq M_N(R)$.

$\sigma(a)$ is the transpose of the matrix of $v \mapsto va$ on I in the basis f; so this is "$A \simeq \mathrm{End}(I)^{op} \simeq M_N(R)$".

Proof. That σ is a homomorphism with contractible fibres is a sheaf proposition (the carriers are sheaves), so by $\|A = M_N(R)\|_T$ we may take $A = M_N(R)$. Then $I = \delta[x]$ with x_k invertible, by Lemma 5.3, and $f(e_i) = xy_i^T$. Put $D_{jk} = g(xe_k^T)_j$. Since $w \mapsto g(xw^T)_j$ is linear, $\sigma(M) = YMZ$ with $Y_{im} = (y_i)_m$ and $Z_{kj} = D_{jk}$. Moreover $YZ = 1$, since $(YZ)_{ij} = g(xy_i^T)_j = g(f(e_i))_j$. By Lemma 3.2, $ZY = 1$, so σ is conjugation, an automorphism. □

Proof of Theorem 1.2. The goal is a proposition. By Proposition 6.3, $X = \mathrm{RI}(A)$ for some $A : \mathrm{AZ}_n$, so a point of X is a right ideal $I : \mathrm{RI}(A)$. By (H4) we may take a basis of I. Lemma 7.1 gives $A \simeq M_N(R)$, hence $A = M_N(R)$, and $X = \mathrm{RI}(A) = \mathrm{RI}(M_N(R)) = \mathbb{P}^n$ by Lemma 5.3. □

8. The étale topology

For $f = X^{d+1} + a_dX^d + \cdots + a_0$ monic of positive degree, let $\mathrm{Roots}(f)$ be the type of $x : R$ with $f(x) = 0$. It is $\mathrm{Spec}(R[X]/(f))$, and $R[X]/(f)$ is free with basis $1, X, \ldots, X^d$. The *étale topology* is given by the propositions $G_f = \|\mathrm{Roots}(f)\|$, indexed by the coefficients $(d, a_0, \ldots, a_d)$ (sevb 1.1.1–1.1.2).

Lemma 8.1. *A set X is a sheaf if its identity types are sheaves and every map $G_f \to X$ is merely constant. A set that is merely a sheaf is a sheaf.*

Proof. For $h : G_f \to X$, "h is constant with value x" determines x, since identity types are sheaves; so the mere value is a value. The second claim follows from the first. □

Lemma 8.2 (sevb 1.1.3, 1.1.4)**.** *R is a sheaf, and so is every affine scheme.*

Proof. Let $B = R[X]/(f)$. By synthetic quasi-coherence, $B \to (\mathrm{Roots}(f) \to R)$ is a bijection. Given $h : G_f \to R$, let $a \in B$ correspond to $x \mapsto h(x)$. For a root x, this function is constant with value $h(x)$, so $a = h(x) \cdot 1$. Hence the constant coefficient r of a satisfies $r = h(x)$ for every root, i.e. $r = h(g)$ for all $g : G_f$. Uniqueness: if $G_f \to r = s$, then $r \cdot 1$ and $s \cdot 1$ agree at every root, so they are equal in B, and $r = s$ since 1 is part of a basis. Affine schemes are solution sets of equations, so they are sheaves by Lemma 2.1. □

Lemma 8.3. *Open propositions are sheaves.*

Proof. $\mathrm{Roots}(f)$ is not empty: otherwise $\mathrm{Spec}(B)$ is empty, so $1 = 0$ in B, while $R \to B$ is injective and $1 \neq 0$ in R. So $G_f \to U$ gives $\neg\neg U$, and open propositions are double-negation stable. □

Lemma 8.4 (sevb 1.2.2)**.** *For U an open subset of* $\mathrm{Roots}(f)$*, the proposition $\forall x.\, U(x)$ is open.*

Proof. The goal is a proposition, and merely there are $h_1, \ldots, h_m \in B$ such that $U(x)$ holds exactly when some $h_s(x)$ is invertible. Then $\forall x.\, U(x)$ holds exactly when the h_s generate the unit ideal of B, i.e. when the R-linear map $B^m \to B$, $(b_s) \mapsto \sum_s h_s b_s$, is surjective. In the basis of B this is a matrix over R. Surjectivity of a matrix with $k+1$ columns is equivalent to: for some row j the last entry is invertible and the Schur complement at that entry is surjective. By induction on k this is a finite disjunction of conjunctions of open propositions, hence open. □

This replaces sevb's argument through a closed subscheme of $\mathbb{P}^1$.

Lemma 8.5. *The type of open propositions is a sheaf.*

Proof. Equality of open propositions is logical equivalence, which is a sheaf by Lemma 8.3. Given $h : G_f \to \mathrm{Open}$, the proposition $\forall y : \mathrm{Roots}(f).\, h(y)$ is open by Lemma 8.4, and it is equivalent to $h(g)$ for every $g : G_f$, since G_f is a proposition. Conclude by Lemma 8.1. □

Proposition 8.6 (sevb 1.2.1, 1.2.3)**.** *Every scheme is a sheaf.*

Proof. By Lemma 8.1 we may fix a finite cover of X by open affine charts U_s. *Identity types.* Given $G_f \to x = y$, merely x is in some chart U; "$y \in U$" is open, hence a sheaf, so $y \in U$, and U is affine, hence a sheaf (Lemma 8.2), so $x = y$. *Maps.* Let $h : G_f \to X$. For each s, $g \mapsto (h(g) \in U_s)$ is a map into open propositions, hence constant with some value V_s (Lemma 8.5). Since the charts cover, $G_f \to \bigvee_s V_s$, a finite disjunction of opens, hence a sheaf, so $\bigvee_s V_s$ holds. For s with V_s, h lands in the sheaf U_s, so it is constant. □

Corollary 8.7. *For the étale topology, Theorem 1.2 holds under (H2) and (H3).*

(H2) and (H3) hold for localisation at a family of propositions [4]; they are not formalised here. (H4) is the use made of sevb's Proposition 1.3.5; it is proved in [3] for algebras with $\|A = M_N(R)\|_T$, which is all that the proof of Theorem 1.2 uses, without any theory of flatness.

9. Correspondence with the formalisation

Everything is in `sag/chatelet`: about 4000 lines in 46 files, on top of the library for synthetic algebraic geometry and projective space, which is included in the same directory; the descent files `Descent*` are described in [3]. The tree of 393 files checks in 11 s in one process. Every file checks in under 0.2 s, except `Prop331Str` (2 s). The hypotheses are module parameters, never postulates: `sag` (the SAG axioms), `sh` (H2 and H3: the record `SheafClosure.Ops`, with field `lex`), `R-modal` (H1), `descends` (H4), the last one discharged for the étale topology by `ChateletEtaleDescent`. Functional extensionality is derived from univalence. Large types are handled by universe levels: $\mathrm{RI}(A)$ and AZ_n are one level up, SB_n two levels up.

Statement	**File and name**
Sheaves, $\|-\|_T$ (H2), (H3)	`SheafTopology.isModal`, `SheafClosure.Ops`
Lemma 2.1	`SheafLemmas`: `modal-Pi`, `modal-Sigma`, `modal-Id`, `modal-retract`, `modal-of-sec`
Lemma 2.2	`AzumayaProbe`: `PropT`, `modal-PropT`, `propT-ext`
Lemma 2.3	`Deloop.Lemma.theorem`, `DeloopLevels.deloop`
Lemmas 3.1, 3.2	`MatrixLeftInverse`: `pivot`, `k2`
$M_N(R)$, E_{ij}	`MnAlgebra.algebra`, `MnElementary.E`
Proposition 4.1	`SkolemNoether.Endo.theorem`
Corollary 4.2	`SkolemNoetherPGL`: `alpha`, `equivalence`, `equivalence-iso`
β	`AutPGL.aut-of`, `corollary-5-6`; instances in `AutPn`
Definition 5.1, AZ_n, SB_n	`AzumayaProbe`: `Ideal`, `Basis`, `Retraction`, `RI`, `RI-modal`, `Types.AZ`, `Types.SB`
Lemma 5.3, sevb 2.4.3	`Delta`: `Col`, `Memb`, `delta-map`, `inj`, `fib-point`, `delta`, `lemma-2-4-3`
Structure identity principle	`AlgSIP.sip`, `AlgSIPInd.iso-ind`
Lemma 6.1	`TriangleRI`: `RIm.map`, `transport`
Lemma 6.2	`Triangle`: `OfA.tri-RI`, `tri`, `omega`

Proposition 6.3	`Prop331.prop-3-3-1`; algebra structures: `Prop331Str.str-modal`; isEquiv is a proposition: `Prop331Paths.isEquiv-prop`
Lemma 7.1	`Lemma341.lemma-3-4-1`
Theorem 1.2	`Theorem342.theorem-3-4-2`
Étale topology	`EtaleTopology`: `I`, `Root`, `G`
Lemma 8.1	`EtaleModal`: `Separated.separated-factor`, `Mere.modal`
Lemma 8.2	`EtaleRing`: `R-local`, `affine-local`; `EtaleRoots.spec-root`
Lemma 8.3	`EtaleOpen.open-local`; `EtaleRoots`: `not-not-root`
Lemma 8.4	`EtaleCompact.At.compact`; `EtaleIdealOpen.unimodular-open`, `EtaleSurj.surj-open`
Lemma 8.5	`EtaleOpenType.opens-local`
Proposition 8.6	`EtaleScheme.etale-scheme`
Corollary (étale)	`ChateletEtale.At.chatelet`
Corollary, (H4) proved	`ChateletEtaleDescent.At.chatelet`

10. Remarks on sevb

The numbering is that of sevb. Sections 2.2, 2.3 and 2.5 of sevb have not been formalised; Section 1.3 is replaced by [3].

Gaps.

(1) *Lemma 2.4.2 (surjectivity) and Proposition 3.3.1: "same dimension, hence equal".* Over a ring a free submodule of the same rank need not be the whole module: $(t) \subseteq R$ is free of rank one when t is a non-zero-divisor. The step holds when the inclusion has a left inverse, by Lemma 3.2. So Gr_k should consist of split subspaces (frames with a left inverse), which is what "matrices of rank k" in the proof of Lemma 2.1.4 should mean. In Proposition 3.3.1 no dimension argument is needed: the inclusion applied to P^{-1} gives the other inclusion.

(2) *Theorem 3.4.2, "we merely have that $I = R^{n+1}$".* From $I : \mathrm{RI}(A)$ one only has $\|I = R^{n+1}\|_T$. Passing to $\|A = M_{n+1}(R)\|$ uses that "finite free" is a sheaf proposition (Proposition 1.3.5, or the hypothesis that the type of finite free modules is a sheaf). This is the only use of descent in the main line; it should be explicit.

(3) *Definition 1.1.1, Remark 1.1.2: the degree of g must be positive.* For $g = 1$, $\mathrm{Spec}(R[X]/g)$ is empty, and only contractible types are local. The proof of Lemma 1.1.3 fails for $n = 0$ (the equaliser of $R^0 \rightrightarrows R^0 \otimes R^0$ is 0), and "$\neg\neg\mathrm{Spec}(R[X]/g)$" in Proposition 1.2.3 is false. For positive degree it holds since an empty spectrum gives $1 = 0$ in $R[X]/g$, while $R \to R[X]/g$ is injective; this deserves a line.

Imprecisions and simplifications.

- *Lemma 2.1.5, "we can proceed as if R was a field".* Extending a basis of I to a basis of A presupposes a complement. With a left inverse g of the frame f, "$x \in I$" is the closed condition $f(gx) = x$.
- *Lemma 3.2.2, "from $V_0^{n+1} = R^{n+1}$ we get $\|V_0 = R\|$".* This uses uniqueness of rank, which is not stated, after Lemma 3.2.1. The detour through projective modules can be avoided (Proposition 4.1): by locality one finds j and w in the image of e_{00} with a linear form μ, $\mu(w) = 1$; then $c \mapsto \sum_k c_k e_{k0} w$ has an explicit left inverse, so it is

invertible by Lemma 3.2, and it intertwines e_{ij} and E_{ij}. Neither Lemma 3.2.1 nor ranks are needed.

- *Proposition 3.2.3 and Lemma 2.4.2.* "$\lambda \neq 0$" should be "λ a unit" (equivalent in synthetic algebraic geometry, but this should be said).
- *Proposition 3.3.1.* The formula for $\delta(Y)$ is garbled (Y versus X, M_n, and the products do not type check). With $\delta[x] = \{xy^T\}$ the triangle is the equality of sets $P\,\delta[x]\,P^{-1} = \delta[Px]$, since $P(xy^T)P^{-1} = (Px)(P^{-T}y)^T$. "$\mathrm{ap}_{\mathrm{RI}}(\alpha(P))$" should be transport along $\mathrm{ap}_{\mathrm{RI}}(\mathrm{sip}(\alpha P))$; that this is $I \mapsto \varphi[I]$ is a lemma (Lemma 6.1). Lemma 3.1.2 needs AZ_n and SB_n to be connected sheaves, which is not checked: SB_n is a sheaf by lexness, and AZ_n because algebra structures on a sheaf form a sheaf. The identifications $\Omega\mathrm{AZ}_n = \mathrm{Aut}(M_{n+1}(R))$ (structure identity principle) and $\Omega\mathrm{SB}_n = \mathrm{Aut}(\mathbb{P}^n)$ (univalence) are implicit.
- *Lemma 3.4.1.* "Sends M to $\delta^{-1} \circ M \circ \delta$" should be $\theta^{-1} \circ M^T \circ \theta$.
- *Lemma 1.1.3.* A shorter proof: by synthetic quasi-coherence $R[X]/g \simeq (\mathrm{Spec}(R[X]/g) \to R)$, and a map $\|\mathrm{Spec}\| \to R$ corresponds to an element equal to $h(x) \cdot 1$, whose constant coefficient is the value (Lemma 8.2).
- *Lemma 1.2.2.* An algebraic proof, without $\mathbb{P}^1$, is Lemma 8.4.
- *Hypotheses of Section 2.* With right ideals valued in sheaf propositions, the main line uses only: R is a sheaf, lexness, and descent of finite freeness. That $\mathbb{P}^n$ is a sheaf then follows from Lemma 5.3. "Schemes are sheaves" is used only for Remark 3.3.2.
- *Lemma 2.5.2.* "We can assume $x_0 \neq 0$" needs a case split on which coordinate of the point is invertible, not only $C(a, b) = C(1/a, -b/a)$.

Typos. $M_n(R)$ for $M_{n+1}(R)$ in Lemma 2.4.2 (twice), Proposition 3.3.1 (the triangle and $\delta(Y)$) and Lemma 3.4.1; Lemma 1.2.1, "$f^{-1}(U_i)$ is an in P" (open); Lemma 2.5.2, "$ay_0 + bz_0 = 1$" for $ay_0^2 + bz_0^2 = 1$, and "ϕ inverse to ϕ" for "inverse to ψ"; "generalise of", "over over", "comuutative" (introduction), "when when" (2.4.2), "an well-known" (3.2), "Let us proof" (2.5.2), "conclude than" (3.3.2).

References

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